**Data Supplement**

# Visit-to-visit blood pressure variability and cardiovascular outcomes in patients receiving intensive versus standard blood pressure control: Insights from the STEP trial

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**Table S1. Baseline Characteristics of Participants by Median of Indexes of Systolic Blood Pressure Variability**

| Characteristics | CV | | SD | | delta | | ARV | | VIM | |
|---|---|---|---|---|---|---|---|---|---|---|
| | <=6.26 | >6.26 | <=5.44 | >5.44 | <=15 | >15 | <=0.04 | >0.04 | <=10.79 | >10.79 |
| Number (%) | 3839 | 3839 | 3836 | 3842 | 3939 | 3739 | 3698 | 3980 | 3840 | 3838 |
| Age, mean (SD) | 66.21(4.82) | 66.16(4.76) | 66.21(4.81) | 66.16(4.78) | 66.23(4.83) | 66.13(4.75) | 66.26(4.8) | 66.11(4.79) | 66.26(4.81) | 66.11(4.78) |
| Gender, male，n (%) | 2087(54.36) | 2023(52.7) | 2059(53.68) | 2051(53.38) | 2123(53.9) | 1987(53.14) | 1991(53.84) | 2119(53.24) | 2070(53.91) | 2040(53.15) |
| BMI, mean (SD) | 25.51(3.1) | 25.66(3.23)ǂ | 25.52(3.11) | 25.65(3.21) | 25.52(3.1) | 25.65(3.23) | 25.54(3.12) | 25.63(3.21) | 25.54(3.11) | 25.63(3.22) |
| Systolic blood pressure, mean (SD) | 130.69(7.58) | 133.81(9.05)* | 130.72(7.62) | 133.78(9.03)* | 130.76(7.66) | 133.83(9.02)* | 131.25(7.81) | 133.19(8.98)* | 131.19(7.8) | 133.32(9)* |
| Diastolic blood pressure, mean (SD) | 77.96(5.64) | 77.94(6.57) | 78.04(5.68) | 77.86(6.53) | 78.01(5.7) | 77.89(6.53) | 78.16(5.73) | 77.76(6.45)† | 78.14(5.73) | 77.76(6.48)† |
| Current smoking, n (%) | 587(15.29) | 642(16.72) | 573(14.94) | 656(17.07)ǂ | 585(14.85) | 644(17.22)† | 550(14.87) | 679(17.06)† | 571(14.87) | 658(17.14)† |
| Medication history, n (%) | | | | | | | | | | |
| Diabetes mellitus | 664(17.3) | 785(20.45)* | 664(17.31) | 785(20.43)* | 694(17.62) | 755(20.19)† | 638(17.25) | 811(20.38)* | 671(17.47) | 778(20.27)† |
| Dyslipidemia | 1285(33.47) | 1513(39.41)* | 1274(33.21) | 1524(39.67)* | 1318(33.46) | 1480(39.58)* | 1209(32.69) | 1589(39.92)* | 1267(32.99) | 1531(39.89)* |
| CVD History | 621(16.18) | 715(18.62)† | 622(16.21) | 714(18.58)† | 636(16.15) | 700(18.72)† | 592(16.01) | 744(18.69)† | 625(16.28) | 711(18.53)† |
| Chronic kidney dysfunction | 51(1.33) | 51(1.33) | 47(1.23) | 55(1.43) | 52(1.32) | 50(1.34) | 46(1.24) | 56(1.41) | 47(1.22) | 55(1.43) |
| eGFR, | 109.21(23. | 108.2(22.7 | 109(23.26 | 108.4(22.9 | 108.99(23. | 108.4(22.9 | 108.94(23. | 108.48(22. | 108.93(23. | 108.48(22. |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| mean (SD) | 44) | 2) | ) | 1) | 25) | 1) | 35) | 84) | 29) | 89) |
| Antihypertensive Medications, n (%) | | | | | | | | | | |
| ARB | 2606(67.88) | 2700(70.33)ǂ | 2575(67.13) | 2731(71.08)* | 2651(67.3) | 2655(71.01)* | 2487(67.25) | 2819(70.83)* | 2582(67.24) | 2724(70.97)* |
| CCB | 3039(79.16) | 3038(79.14) | 3040(79.25) | 3037(79.05) | 3114(79.06) | 2963(79.25) | 2936(79.39) | 3141(78.92) | 3050(79.43) | 3027(78.87) |
| Diuretic | 297(7.74) | 383(9.98)* | 279(7.27) | 401(10.44)* | 292(7.41) | 388(10.38)* | 263(7.11) | 417(10.48)* | 275(7.16) | 405(10.55)* |
| β-blockers | 94(2.45) | 177(4.61)* | 87(2.27) | 184(4.79)* | 88(2.23) | 183(4.89)* | 83(2.24) | 188(4.72)* | 88(2.29) | 183(4.77)* |
| Number of antihypertensive medication classes | | | | | | | | | | |
| 0 | 88(2.29) | 136(3.54)* | 92(2.4) | 132(3.44)* | 94(2.39) | 130(3.48)* | 83(2.24) | 141(3.54)* | 88(2.29) | 136(3.54)* |
| 1 | 1739(45.3) | 1514(39.44)* | 1747(45.54) | 1506(39.2)* | 1796(45.6) | 1457(38.97)* | 1689(45.67) | 1564(39.3)* | 1748(45.52) | 1505(39.21)* |
| 2 | 1726(44.96) | 1778(46.31)* | 1737(45.28) | 1767(45.99)* | 1774(45.04) | 1730(46.27)* | 1680(45.43) | 1824(45.83)* | 1746(45.47) | 1758(45.81)* |
| ≥3 | 286(7.45) | 411(10.71)* | 260(6.78) | 437(11.37)* | 275(6.98) | 422(11.29)* | 246(6.65) | 451(11.33)* | 258(6.72) | 439(11.44)* |
| Other medications | | | | | | | | | | |
| Aspirin | 273(7.11) | 378(9.85)* | 268(6.99) | 383(9.97)* | 274(6.96) | 377(10.08)* | 262(7.08) | 389(9.77)* | 274(7.14) | 377(9.82)* |
| Statin | 664(17.3) | 794(20.68)* | 640(16.68) | 818(21.29)* | 669(16.98) | 789(21.1)* | 603(16.31) | 855(21.48)* | 635(16.54) | 823(21.44)* |
| Antidiabetic agent | 563(14.67) | 693(18.05)* | 562(14.65) | 694(18.06)* | 588(14.93) | 668(17.87)* | 544(14.71) | 712(17.89)* | 568(14.79) | 688(17.93)* |
| Framingham score ≥15 | 2912(75.85) | 3084(80.33)* | 2936(76.54) | 3060(79.65)* | 3019(76.64) | 2977(79.62)† | 2846(76.96) | 3150(79.15)ǂ | 2953(76.9) | 3043(79.29)ǂ |

BMI is the weight in kilograms divided by the square of the height in meters. Chronic kidney dysfunction was defined as an estimated glomerular filtration rate of less than 60 ml per minute per 1.73 $m^2$. eGFR was calculated according to the 4-component Modification of Diet in Renal Disease study prediction equation. A Framingham Risk Score of 15% or higher indicates a high 10-year risk of cardiovascular disease. BMI: body mass index; ARB, angiotensin receptor blocker; CCB, calcium channel blocker; eGFR, glomerular filtration rate estimated from the serum creatinine; CVD, cardiovascular disease; CV, coefficient of variation; ARV, average real variability; VIM, variability independent of the mean.

Significance of the between-group differences: *P≤0.001; †P≤0.05; and ‡P≤0.01.

**Table S2. Baseline Characteristic of Participants**

| Characteristic | Total | Intensive treatment group | Standard treatment group | P value |
|---|---|---|---|---|
| Number | 7678 | 3875 | 3803 | |
| Age, mean (SD) | 66.19(4.79) | 66.16 (4.84) | 66.21 (4.75) | 0.671 |
| Gender, male，n (%) | 4110(53.53) | 2057 (53.1) | 2053 (54.0) | 0.429 |
| BMI, mean (SD) | 25.58(3.16) | 25.54 (3.16) | 25.63 (3.17) | 0.213 |
| Systolic blood pressure, mean (SD) | 132.25(8.49) | 128.41 (7.67) | 136.17 (7.43) | <0.001 |
| Diastolic blood pressure, mean (SD) | 77.95(6.12) | 76.72 (5.95) | 79.20 (6.04) | <0.001 |
| Current smoking, n (%) | 1229(16.01) | 628 (16.2) | 601 (15.8) | 0.63 |
| Medication history n (%) | | | | |
| Diabetes mellitus | 1449(18.87) | 726 (18.7) | 723 (19.0) | 0.758 |
| Dyslipidemia | 2798(36.44) | 1440 (37.2) | 1358 (35.7) | 0.186 |
| CVD History | 1336(17.4) | 677 (17.6) | 659 (17.5) | 0.869 |
| Chronic kidney dysfunction | 102(1.33) | 50(1.29) | 52(1.37) | 0.768 |
| eGFR, mean (SD) | 108.7(23.09) | 109.18(23.09) | 108.21(23.07) | 0.066 |
| Antihypertensive medications, n (%) | | | | |
| ARB | 5306(69.11) | 2754 (71.1) | 2327 (61.2) | <0.001 |
| CCB | 6077(79.15) | 3056 (78.9) | 2768 (72.8) | <0.001 |
| Diuretic | 680(8.86) | 473(12.21) | 207(5.44) | <0.001 |
| β-blockers | 271(3.53) | 156(4.03) | 115(3.02) | 0.017 |
| Number of antihypertensive medication classes | | | | <0.001 |
| 0 | 224(2.92) | 92(2.37) | 132(3.47) | |
| 1 | 3253(42.37) | 1334(34.43) | 1919(50.46) | |
| 2 | 3504(45.64) | 1966(50.74) | 1538(40.44) | |
| ≥3 | 697(9.08) | 483(12.46) | 214(5.63) | |
| Other medications, n (%) | | | | |
| Aspirin | 651(8.48) | 321 (8.3) | 330 (8.7) | 0.536 |
| Statin | 1458(18.99) | 737 (19.0) | 721 (19.0) | 0.946 |
| Antidiabetic agent, n (%) | 1256(16.36) | 616 (15.9) | 640 (16.8) | 0.27 |
| FRS ≥15%, n (%) | 5996(78.09) | 3036 (78.3) | 2960 (77.8) | 0.585 |
| BPV | | | | |
| CV_SBP | 0.05(0.03) | 0.05(0.03) | 0.05(0.03) | <0.001 |
| SD_SBP | 6.31(4.03) | 6.48(4.18) | 6.14(3.87) | <0.001 |
| Delta_SBP | 17.26(11.1) | 17.69(11.43) | 16.83(10.73) | <0.001 |
| ARV_SBP | 7.54(5.22) | 7.71(5.41) | 7.36(5.01) | 0.003 |
| VIM_SBP | 12.42(7.76) | 12.99(8.08) | 11.84(7.37) | <0.001 |
| CV_DBP | 0.06(0.03) | 0.06(0.03) | 0.06(0.03) | <0.001 |
| SD_DBP | 4.48(2.26) | 4.55(2.32) | 4.42(2.19) | 0.015 |
| Delta_DBP | 12.2(6.16) | 12.35(6.32) | 12.04(5.98) | 0.024 |
| ARV_DBP | 5.37(3.05) | 5.44(3.19) | 5.29(2.89) | 0.031 |

| | | | | |
|---|---|---|---|---|
| VIM_DBP | 9.47(4.77) | 9.6(4.91) | 9.33(4.62) | 0.012 |

BMI body mass index, CVD cardiovascular disease, ARB angiotensin receptor blocker, CCB calcium channel blocker, FRS Framingham risk score, eGFR estimated glomerular filtration rate, BPV blood pressure variability, CV coefficient variability, SD standard deviation, ARV average real variability, VIM variability independent of the mean of blood pressure.

**Table S3. Associations of diastolic BPV with Primary End Point, MACE and ACS in the standard treatment group using systolic BP-adjusted cox regression model**

| BPV index | Primary | | MACE | | ACS | |
|---|---|---|---|---|---|---|
| | HR (95%CI) | P | HR (95%CI) | P | HR (95%CI) | P |
| CV | 1.23 [1.05, 1.45] | 0.011 | 1.21 [1.01, 1.46] | 0.037 | 1.27 [1.04, 1.55] | 0.018 |
| SD | 1.25 [1.06, 1.47] | 0.007 | 1.23 [1.02, 1.47] | 0.027 | 1.29 [1.06, 1.57] | 0.012 |
| Delta | 1.28 [1.09, 1.51] | 0.002 | 1.25 [1.04, 1.50] | 0.017 | 1.32 [1.09, 1.62] | 0.006 |
| ARV | 1.20 [1.01, 1.42] | 0.041 | 1.20 [0.99, 1.45] | 0.065 | 1.21 [0.97, 1.50] | 0.089 |
| VIM | 1.25 [1.06, 1.47] | 0.007 | 1.23 [1.02, 1.47] | 0.027 | 1.29 [1.06, 1.57] | 0.012 |

Hazard ratios (95% CI) express the difference in primary end point，MACE and ACS associated with 1-SD increment in visit-to-visit variability indexes in the standard treatment group. Adjusted model accounted for mean systolic blood pressure during the follow-up period, sex，age，body mass index，baseline CKD status，diabetes，hyperlipidemia， cardiovascular disease，current smoking, the number of antihypertensive medication classes received at the 6-month study visit，use of antihypertensive medications by drug class, that is, diuretics, β-blockers, inhibitors of the renin-angiotensin, calcium-channel blockers at 6-month，baseline intake of aspirin, lipid-lowering drugs, and antidiabetic agents. BPV, blood pressure variability; CV, coefficient of variation; ARV average real variability; VIM, variability independent of the mean.

**Table S4. Associations of diastolic BPV with Primary End Point, MACE and ACS in the intensive treatment group using systolic BP-adjusted cox regression model**

| BPV index | Primary | | MACE | | ACS | |
|---|---|---|---|---|---|---|
| | HR (95%CI) | P | HR (95%CI) | P | HR (95%CI) | P |
| CV | 1.10 [0.90, 1.33] | 0.349 | 1.14 [0.93, 1.41] | 0.209 | 1.13 [0.90, 1.41] | 0.295 |
| SD | 1.08 [0.89, 1.32] | 0.424 | 1.13 [0.91, 1.39] | 0.269 | 1.12 [0.89, 1.40] | 0.35 |
| Delta | 1.09 [0.90, 1.32] | 0.391 | 1.12 [0.91, 1.39] | 0.279 | 1.11 [0.88, 1.40] | 0.364 |
| ARV | 0.99 [0.80, 1.21] | 0.904 | 0.98 [0.78, 1.22] | 0.828 | 0.97 [0.76, 1.24] | 0.793 |
| VIM | 1.08 [0.89, 1.32] | 0.421 | 1.13 [0.91, 1.39] | 0.266 | 1.12 [0.89, 1.41] | 0.348 |

Hazard ratios (95% CI) express the difference in primary end point，MACE and ACS associated with 1-SD increment in visit-to-visit variability indexes in the standard treatment group. Adjusted model accounted for mean systolic blood pressure during the follow-up period, sex, age，body mass index，baseline CKD status，diabetes, hyperlipidemia, cardiovascular disease, current smoking, the number of antihypertensive medication classes received at the 6-month study visit，use of antihypertensive medications by drug class, that is, diuretics, β-blockers, inhibitors of the renin-angiotensin, calcium-channel blockers at 6-month，baseline intake of aspirin, lipid-lowering drugs, and antidiabetic agents. BPV, blood pressure variability; CV, coefficient of variation; ARV average real variability; VIM, variability independent of the mean.

**Table S5 Sensitivity analysis using the specified alternative methods to calculate visit-to-visit BPV. Association of BPV with outcomes in the standard group**

| Variability model | Primary | | MACE | | ACS | |
|---|---|---|---|---|---|---|
| | HR (95%CI) | P value | HR (95%CI) | P value | HR (95%CI) | P value |
| Systolic BPV | | | | | | |
| Unadjusted models | | | | | | |
| CV | 1.17 [0.98, 1.40] | 0.088 | 1.18 [0.97, 1.44] | 0.099 | 1.19 [0.95, 1.49] | 0.129 |
| SD | 1.19 [1.00, 1.42] | 0.054 | 1.19 [0.98, 1.44] | 0.082 | 1.20 [0.96, 1.50] | 0.107 |
| Delta | 1.20 [1.01, 1.43] | 0.040 | 1.20 [0.99, 1.45] | 0.064 | 1.21 [0.97, 1.50] | 0.093 |
| ARV | 1.18 [0.99, 1.40] | 0.065 | 1.16 [0.96, 1.41] | 0.128 | 1.12 [0.90, 1.41] | 0.315 |
| VIM | 1.17 [0.98, 1.40] | 0.080 | 1.18 [0.97, 1.44] | 0.096 | 1.19 [0.95, 1.49] | 0.124 |
| Adjusted models | | | | | | |
| CV | 1.16 [0.97, 1.39] | 0.111 | 1.18 [0.97, 1.44] | 0.098 | 1.19 [0.95, 1.50] | 0.120 |
| SD | 1.15 [0.96, 1.38] | 0.121 | 1.17 [0.96, 1.43] | 0.114 | 1.18 [0.95, 1.48] | 0.141 |
| Delta | 1.16 [0.97, 1.39] | 0.100 | 1.18 [0.97, 1.45] | 0.095 | 1.19 [0.95, 1.49] | 0.127 |
| ARV | 1.14 [0.95, 1.37] | 0.149 | 1.15 [0.94, 1.40] | 0.182 | 1.10 [0.87, 1.39] | 0.414 |
| VIM | 1.16 [0.97, 1.39] | 0.112 | 1.18 [0.97, 1.44] | 0.101 | 1.19 [0.95, 1.49] | 0.124 |
| Diastolic BPV | | | | | | |
| Unadjusted models | | | | | | |
| CV | 1.38 [1.17, 1.62] | <0.001 | 1.36 [1.13, 1.63] | 0.001 | 1.41 [1.16, 1.73] | 0.001 |
| SD | 1.39 [1.18, 1.63] | <0.001 | 1.36 [1.13, 1.63] | 0.001 | 1.42 [1.16, 1.73] | 0.001 |
| Delta | 1.42 [1.21, 1.66] | <0.001 | 1.40 [1.17, 1.67] | <0.001 | 1.48 [1.22, 1.79] | <0.001 |
| ARV | 1.27 [1.07, 1.51] | 0.005 | 1.30 [1.07, 1.56] | 0.007 | 1.29 [1.04, 1.60] | 0.020 |
| VIM | 1.39 [1.18, 1.63] | <0.001 | 1.36 [1.13, 1.63] | 0.001 | 1.42 [1.16, 1.73] | 0.001 |
| Adjusted models | | | | | | |
| CV | 1.37 [1.16, 1.61] | <0.001 | 1.35 [1.12, 1.62] | 0.002 | 1.43 [1.16, 1.75] | 0.001 |
| SD | 1.36 [1.16, 1.60] | <0.001 | 1.34 [1.12, 1.61] | 0.002 | 1.41 [1.16, 1.73] | 0.001 |
| Delta | 1.40 [1.20, 1.65] | <0.001 | 1.39 [1.16, 1.67] | <0.001 | 1.49 [1.22, 1.81] | <0.001 |
| ARV | 1.28 [1.07, 1.52] | 0.006 | 1.30 [1.08, 1.58] | 0.007 | 1.32 [1.06, 1.64] | 0.014 |
| VIM | 1.36 [1.16, 1.60] | <0.001 | 1.34 [1.12, 1.61] | 0.002 | 1.41 [1.16, 1.73] | 0.001 |

Hazard ratios (95% CI) express the difference in primary end point，MACE and ACS associated with 1-SD increment in visit-to-visit variability indexes in the standard treatment group. Adjusted models accounted for sex，age，body mass index，baseline CKD status，diabetes，hyperlipidemia， cardiovascular disease，current smoking, mean corresponding blood pressure during the follow-up period, the number of antihypertensive medication classes received at the 6-month study visit，use of antihypertensive medications by drug class, that is, diuretics, β-blockers, inhibitors of the renin-angiotensin, calcium-channel blockers at 6-month，baseline intake of aspirin, lipid-lowering drugs, and antidiabetic agents. BPV, blood pressure variability; CV, coefficient of variation; ARV average real variability; VIM, variability independent of the mean.

**Table S6 Sensitivity analysis using the specified alternative methods to calculate visit-to-visit BPV. Association of BPV with outcomes in the intensive group**

| Variability model | Primary | | MACE | | ACS | |
|---|---|---|---|---|---|---|
| | HR (95%CI) | P value | HR (95%CI) | P value | HR (95%CI) | P value |
| Systolic BPV | | | | | | |
| Unadjusted models | | | | | | |
| CV | 0.86 [0.55, 1.36] | 0.532 | 1.21 [0.97, 1.51] | 0.096 | 1.20 [0.94, 1.53] | 0.137 |
| SD | 0.87 [0.55, 1.36] | 0.538 | 1.20 [0.96, 1.49] | 0.104 | 1.19 [0.94, 1.51] | 0.153 |
| Delta | 0.86 [0.55, 1.37] | 0.531 | 1.22 [0.99, 1.52] | 0.068 | 1.21 [0.95, 1.53] | 0.122 |
| ARV | 0.69 [0.40, 1.18] | 0.177 | 1.20 [0.97, 1.48] | 0.102 | 1.13 [0.88, 1.44] | 0.333 |
| VIM | 0.87 [0.55, 1.36] | 0.533 | 1.21 [0.97, 1.51] | 0.097 | 1.20 [0.94, 1.53] | 0.139 |
| Adjusted models | | | | | | |
| CV | 0.78 [0.39, 1.56] | 0.486 | 1.20 [0.93, 1.53] | 0.157 | 1.21 [0.93, 1.59] | 0.159 |
| SD | 0.78 [0.38, 1.59] | 0.488 | 1.20 [0.93, 1.56] | 0.168 | 1.23 [0.92, 1.63] | 0.161 |
| Delta | 0.72 [0.35, 1.49] | 0.378 | 1.23 [0.95, 1.58] | 0.115 | 1.24 [0.94, 1.64] | 0.128 |
| ARV | 0.67 [0.33, 1.40] | 0.290 | 1.16 [0.91, 1.49] | 0.237 | 1.10 [0.83, 1.45] | 0.522 |
| VIM | 0.78 [0.39, 1.57] | 0.487 | 1.20 [0.93, 1.54] | 0.158 | 1.22 [0.93, 1.60] | 0.159 |
| Diastolic BPV | | | | | | |
| Unadjusted models | | | | | | |
| CV | 1.16 [0.95, 1.43] | 0.144 | 1.22 [0.98, 1.53] | 0.077 | 1.23 [0.96, 1.57] | 0.095 |
| SD | 1.15 [0.94, 1.42] | 0.180 | 1.20 [0.96, 1.50] | 0.117 | 1.21 [0.95, 1.55] | 0.129 |
| Delta | 1.14 [0.92, 1.40] | 0.229 | 1.16 [0.92, 1.46] | 0.207 | 1.15 [0.89, 1.47] | 0.277 |
| ARV | 1.09 [0.88, 1.34] | 0.424 | 1.06 [0.84, 1.34] | 0.620 | 1.06 [0.82, 1.36] | 0.675 |
| VIM | 1.15 [0.94, 1.42] | 0.179 | 1.20 [0.96, 1.51] | 0.115 | 1.21 [0.95, 1.55] | 0.128 |
| Adjusted models | | | | | | |
| CV | 1.12 [0.91, 1.38] | 0.289 | 1.16 [0.92, 1.47] | 0.200 | 1.18 [0.91, 1.52] | 0.205 |
| SD | 1.12 [0.91, 1.38] | 0.296 | 1.17 [0.92, 1.47] | 0.195 | 1.18 [0.91, 1.51] | 0.210 |
| Delta | 1.11 [0.90, 1.37] | 0.344 | 1.13 [0.89, 1.43] | 0.307 | 1.12 [0.86, 1.45] | 0.389 |
| ARV | 1.07 [0.86, 1.32] | 0.547 | 1.03 [0.81, 1.32] | 0.781 | 1.03 [0.79, 1.34] | 0.817 |
| VIM | 1.12 [0.91, 1.38] | 0.296 | 1.17 [0.92, 1.47] | 0.195 | 1.18 [0.91, 1.51] | 0.210 |

Hazard ratios (95% CI) express the difference in primary end point, MACE and ACS associated with 1-SD increment in visit-to-visit variability indexes in the intensive treatment group. Adjusted models accounted for sex，age，body mass index，baseline CKD status，diabetes，hyperlipidemia， cardiovascular disease，current smoking, mean corresponding blood pressure during the follow-up period, the number of antihypertensive medication classes received at the 6-month study visit，use of antihypertensive medications by drug class, that is, diuretics, β-blockers, inhibitors of the renin-angiotensin, calcium-channel blockers at

6-month，baseline intake of aspirin, lipid-lowering drugs, and antidiabetic agents. BPV, blood pressure variability; CV, coefficient of variation; ARV average real variability; VIM, variability independent of the mean.

**Table S7. Sensitivity analysis using the specified alternative methods to calculate visit-to-visit BPV. Associations of diastolic BPV with Primary End Point, MACE and ACS in the standard treatment using systolic BP-adjusted cox regression model**

| BPV index | Primary | | MACE | | ACS | |
|---|---|---|---|---|---|---|
| | HR (95%CI) | P | HR (95%CI) | P | HR (95%CI) | |
| CV | 1.34 [1.14, 1.58] | <0.001 | 1.33 [1.11, 1.59] | 0.002 | 1.40 [1.14, 1.71] | 0 |
| SD | 1.35 [1.15, 1.59] | <0.001 | 1.34 [1.11, 1.61] | 0.002 | 1.40 [1.15, 1.71] | 0 |
| Delta | 1.39 [1.18, 1.63] | <0.001 | 1.38 [1.15, 1.66] | <0.001 | 1.47 [1.21, 1.79] | <0 |
| ARV | 1.26 [1.06, 1.50] | 0.01 | 1.30 [1.07, 1.57] | 0.008 | 1.30 [1.05, 1.63] | 0 |
| VIM | 1.35 [1.15, 1.59] | <0.001 | 1.34 [1.11, 1.61] | 0.002 | 1.40 [1.15, 1.71] | 0 |

Hazard ratios (95% CI) express the difference in primary end point，MACE and ACS associated with 1-SD increment in visit-to-visit variability indexes in the standard treatment group. Adjusted model accounted for mean systolic blood pressure during the follow-up period, sex，age，body mass index，baseline CKD status，diabetes，hyperlipidemia， cardiovascular disease，current smoking, the number of antihypertensive medication classes received at the 6-month study visit，use of antihypertensive medications by drug class, that is, diuretics, β-blockers, inhibitors of the renin-angiotensin, calcium-channel blockers at 6-month，baseline intake of aspirin, lipid-lowering drugs, and antidiabetic agents. BPV, blood pressure variability; CV, coefficient of variation; ARV average real variability; VIM, variability independent of the mean.

**Table S8. Sensitivity analysis using the specified alternative methods to calculate visit-to-visit BPV. Associations of diastolic BPV with Primary End Point, MACE and ACS in the intensive treatment using systolic BP-adjusted cox regression model**

| BPV index | Primary | | MACE | | ACS | |
|---|---|---|---|---|---|---|
| | HR (95%CI) | P | HR (95%CI) | P | HR (95%CI) | |
| CV | 1.10 [0.90, 1.36] | 0.351 | 1.17 [0.93, 1.47] | 0.18 | 1.18 [0.92, 1.51] | [illegible] |
| SD | 1.10 [0.89, 1.36] | 0.376 | 1.16 [0.92, 1.47] | 0.2 | 1.18 [0.91, 1.52] | [illegible] |
| Delta | 1.09 [0.88, 1.35] | 0.446 | 1.13 [0.89, 1.43] | 0.317 | 1.12 [0.87, 1.46] | [illegible] |
| ARV | 1.05 [0.84, 1.30] | 0.677 | 1.03 [0.81, 1.32] | 0.798 | 1.03 [0.79, 1.35] | [illegible] |
| VIM | 1.10 [0.89, 1.36] | 0.375 | 1.16 [0.92, 1.47] | 0.199 | 1.18 [0.91, 1.52] | [illegible] |

Hazard ratios (95% CI) express the difference in primary end point，MACE and ACS associated with 1-SD increment in visit-to-visit variability indexes in the standard treatment group. Adjusted model accounted for mean systolic blood pressure during the follow-up period, sex, age，body mass index，baseline CKD status，diabetes, hyperlipidemia, cardiovascular disease, current smoking, the number of antihypertensive medication classes received at the 6-month study visit，use of antihypertensive medications by drug class, that is, diuretics, β-blockers, inhibitors of the renin-angiotensin, calcium-channel blockers at 6-month， baseline intake of aspirin, lipid-lowering drugs, and antidiabetic agents. BPV, blood pressure variability; CV, coefficient of variation; ARV average real variability; VIM, variability independent of the mean.